CS342 · VISUAL PROGRAMMING

Labs Practice

لابات د. حمد الرشيدي · تدريب Output وكتابة كود · محارب

النوع	الأسئلة	من وين
Find Output	Q1 – Q5	Lab 1: Ex 1, 2, 3, 4, 5, 6
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كل الأكواد مجربة، والمخرجات في ملف الحل هي اللي تطلع فعلًا. وأخطاء الكتابة اللي في نص اللاب (مثل liste و Time والفاصلة المنقوطة الناقصة) مصلحة هنا.

Q1 **Division by zero** OUTPUT Lab 1 · Ex 5

(a) What does the program print when y = 0?

```
public class Exemple1 {
    public static void main(String[] args) {
        int x = 10, y = 0;
        int z = x / y;
        System.out.println(x + " / " + y + " = " + z);
        System.out.println("Program End");
    }
}
```

(b) What does it print after adding try / catch?

```
public class Exemple1 {
    public static void main(String[] args) {
        int x = 10, y = 0;
        try {
            int z = x / y;
            System.out.println(x + " / " + y + " = " + z);
        }
        catch (ArithmeticException e) {
            System.out.println("Error: " + e.getMessage());
        }
        System.out.println("Program End");
    }
}
```

Lab 1 · Ex 6 OUTPUT **CalculMoyenne** Q2

What does the program print for each array? Then: which predefined exceptions are caught, and which exceptions were created by the programmer?

(1) {"14", "12", "13", "16", "18", "15"}

(2) {"x", "2", "-1", "30", "18"}

(3) {"x", "-3", "2.3"}

```
class ExceptionEmpty extends Exception {
    public String toString() { return "No note is valid"; }
}
class ExceptionRange extends Exception {
    public ExceptionRange(String s) { super(s); }
}
public class CalculMoyenne {
    private static int average(String[] list) throws ExceptionEmpty {
        int sum = 0, integer, nbNotes = 0;
        for (int i = 0; i < list.length; i++) {
            try {
                integer = Integer.parseInt(list[i]);
                if (integer < 0)
                    throw new ExceptionRange("few");
                else if (integer > 20)
                    throw new ExceptionRange("large");
                else {
                    sum += integer;
                    nbNotes++;
                }
            }
            catch (NumberFormatException nfe) {
                System.out.println("Note no." + (i + 1) + " is not complete");
            }
            catch (ExceptionRange ei) {
                System.out.println("Note no." + (i + 1) + " is too " + ei.getMessage());
            }
        }
        if (nbNotes == 0)
            throw new ExceptionEmpty();
        return sum / nbNotes;
    }
```

```
public static void main(String[] args) {
    String tab[] = { ... };
    try {
        System.out.println("The average is: " + average(tab));
    }
    catch (ExceptionEmpty er) {
        System.out.println(er);
    }
}
```

Lab 1 · Ex 1 OUTPUT **Person, Customer and Owner** Q3

Find the output of the program below.

```
class Person {
    private String lastname, firstname;
    public Person(String l, String f) {
        lastname = l;
        firstname = f;
        System.out.println("Person created");
    }
    public void displaysPerson() {
        System.out.println(lastname + " " + firstname);
    }
}
class Customer extends Person {
    private int numidentity;
    public Customer(String l, String f, int n) {
        super(l, f);
        numidentity = n;
        System.out.println("Customer created");
    }
    public void displaysCustomer() {
        displaysPerson();
        System.out.println("ID: " + numidentity);
    }
}
class Owner extends Person {
    private String nameclub, adrclub;
    public Owner(String l, String f, String n, String a) {
        super(l, f);
        nameclub = n;
        adrclub = a;
        System.out.println("Owner created");
    }
    public void displaysOwner() {
        displaysPerson();
        System.out.println(nameclub + " - " + adrclub);
    }
}
public class Test {
    public static void main(String[] args) {
        Customer c = new Customer("Ali", "Ahmad", 1234);
        c.displaysCustomer();
        Owner o = new Owner("Saad", "Omar", "Star Video", "Buraydah");
        o.displaysOwner();
        Person p = c;
        p.displaysPerson();
```

```
    }
}
```

Lab 1 · Ex 3 OUTPUT **Shapes** Q4

Find the output of the program below.

```
class Point {
    private double x, y;
    public Point(double x, double y) { this.x = x; this.y = y; }
}
abstract class Shape {
    public abstract double perimeter();
    public abstract double area();
}
class Circle extends Shape {
    private Point center;
    private double radius;
    public Circle(Point c, double r) { center = c; radius = r; }
    public void setRadius(double r) { radius = r; }
    public double getRadius() { return radius; }
    public double perimeter() { return 2 * 3.14 * radius; }
    public double area() { return 3.14 * radius * radius; }
}
class Rectangle extends Shape {
    private double length, breadth;
    public Rectangle(double l, double b) { length = l; breadth = b; }
    public void setLength(double l) { length = l; }
    public double perimeter() { return 2 * (length + breadth); }
    public double area() { return length * breadth; }
}
public class TestForm {
    public static void main(String[] args) {
        Shape[] t = { new Rectangle(2, 3), new Circle(new Point(0, 0), 1) };
        for (int i = 0; i < t.length; i++)
            System.out.println(t[i].area() + " " + t[i].perimeter());
        Circle c = (Circle) t[1];
        c.setRadius(2);
        System.out.println(t[1].area());
        ((Rectangle) t[0]).setLength(5);
        System.out.println(t[0].perimeter());
    }
}
```

Lab 1 · Ex 2 – 4 OUTPUT **Compile or not** Q5

Using the lab classes below, say for each line: **OK** (and what it prints), **compile error** or **run-time error**.

```
interface Displayable { void display(); }
interface Shape { double area(); double perimeter(); }
abstract class Person { abstract void writePerson(); }
class Point implements Displayable { ... double getX() ... }   // display() prints "x,y"
class Circle implements Shape, Displayable { ... }              // display() prints "Circle r"
class Rectangle implements Shape, Displayable { ... }           // display() prints "Rect"

(a)  Person p = new Person();
(b)  Shape s = new Shape();
(c)  class Square implements Shape { public double area() { return 1; } }
(d)  Displayable d = new Point(1, 2);  d.display();
(e)  Displayable d = new Point(1, 2);  double v = d.getX();
(f)  Shape s = new Circle(1, new Point(0, 0));  Circle c = (Circle) s;  c.display();
(g)  Shape s = new Circle(1, new Point(0, 0));  Rectangle r = (Rectangle) s;
(h)  Displayable[] t = new Displayable[3];
     t[0] = new Point(1, 2);  t[1] = new Rectangle(2, 3);  t[2] = new Circle(1, (Point) t[0]);
     for (int i = 0; i < 3; i++) t[i].display();
```

Lab 1 · Ex 1 WRITE **Write Person and Customer** Q6

1) Define a Person class with two private attributes: lastname, firstname, a constructor that initializes them, and a method void displaysPerson().

2) Define a Customer class inheriting from Person with a private attribute numidentity, a constructor (lastname, firstname, numidentity) and a method void displaysCustomer().

Lab 1 · Ex 5 WRITE **Three ways to handle division by zero** Q7

Modify the program (x = 10, y = 0) so that it ends well:

1. Using a predefined exception class.
2. Using a predefined exception class with a custom message.
3. Using your own exception class (give a meaningful name and message).

Lab 1 · Ex 7 WRITE **TempsException** Q8

Modify the constructor of the class Temps so that it throws a TempsException (to create) if hours, minutes or seconds do not form a valid time. Change main so that the TempsException is handled by displaying the message "Invalid time".

```
public class Temps {
    private int hours;
    private int minutes;
    private int seconds;

    public Temps(int h, int m, int s) {
        hours = h;
        minutes = m;
        seconds = s;
    }

    public static void main(String[] args) {
        Temps t = new Temps(24, 12, 67);
    }
}
```

Lab 2 · Ex 1 WRITE **MyWindow as a class** Q9

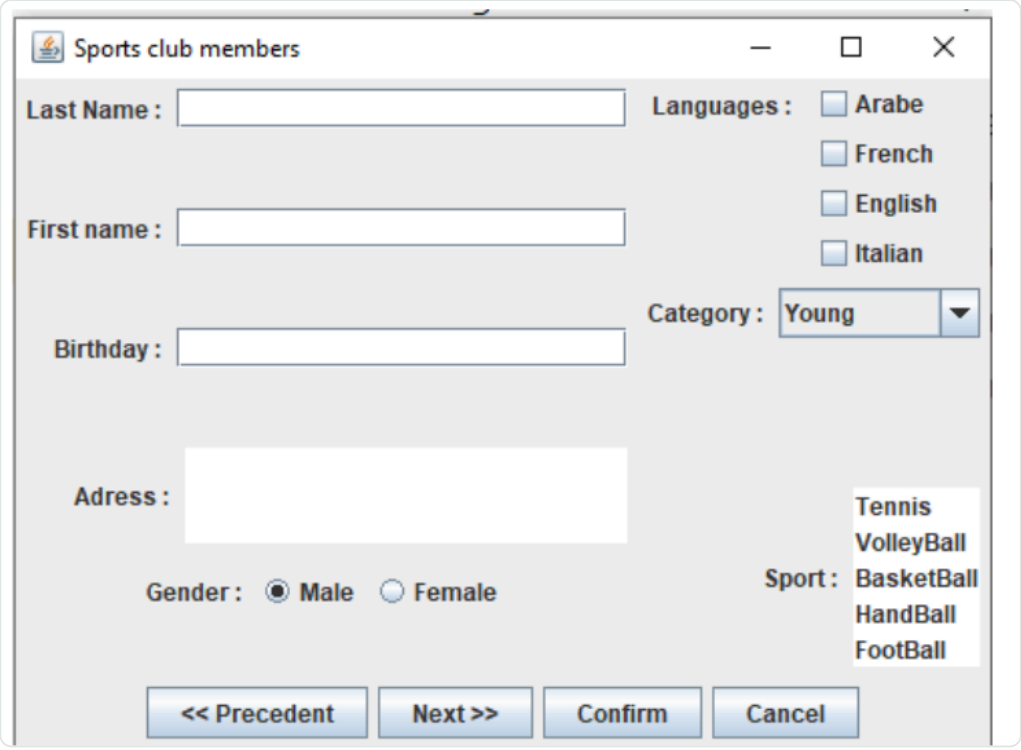
Modify the class MyWindow to get the same GUI when the main program becomes:

`public static void main(String[] args) { MyWindow f = new MyWindow(); }`

```
import javax.swing.*;
public class MyWindow {
    public static void main(String[] args) {
        JFrame f = new JFrame("MyWindow");
        JLabel label = new JLabel("Good Morning");
        JPanel p = (JPanel) f.getContentPane();
        p.add(label);
        f.setSize(300, 200);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

Lab 2 · Ex 2 WRITE **Sports club members (challenge)** Q10

Write the Java program that displays exactly the window below, using JLabel, JTextField, JTextArea, JRadioButton, ButtonGroup, JCheckBox, JComboBox, JList and JButton, and different layout managers.


Sports club members
Last Name :
First name :
Birthday :
Adress :
Gender : Male Female
Languages :
Arabe
French
English
Italian
Category : Young
Sport :
Tennis
VolleyBall
BasketBall
HandBall
FootBall
<< Precedent
Next >>
Confirm
Cancel